CS 412 / 512
Introduction to Computer Graphics

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COMPUTER
SCIENCE

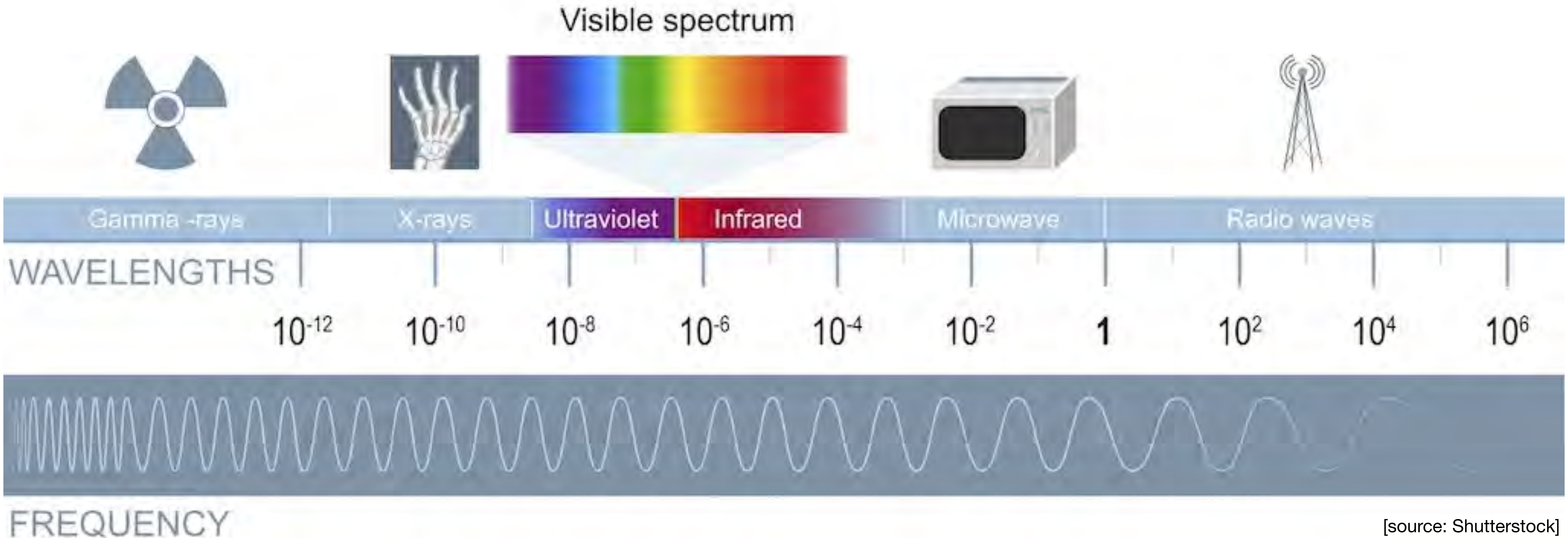

[source: artpartner-images]

[source: numerama]

Light

Visible Light is the part of the electromagnetic spectrum that causes a reaction in our visual systems

The range of visible light wavelengths is roughly 380 - 750 nm



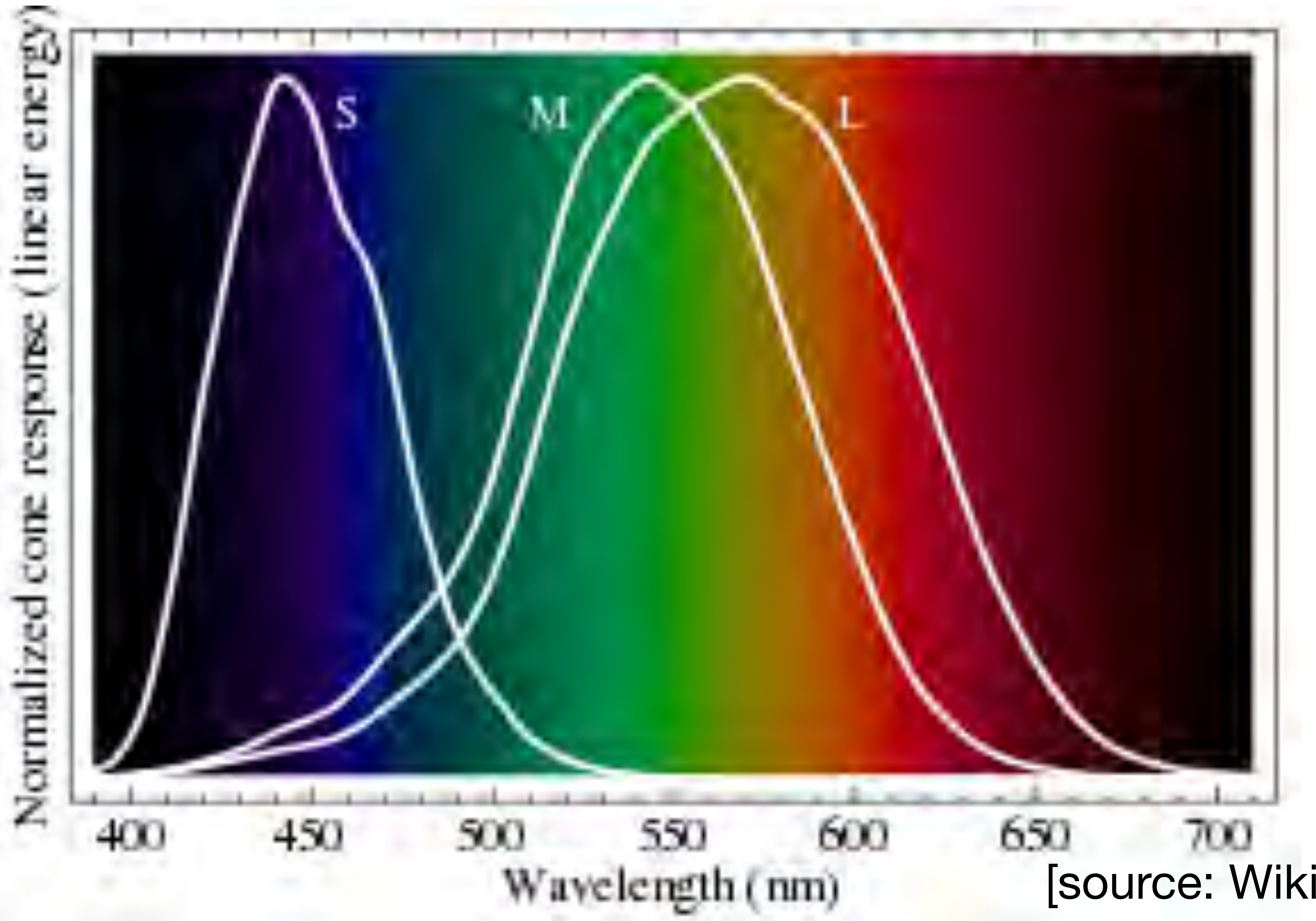
[source: Shutterstock]

Perception

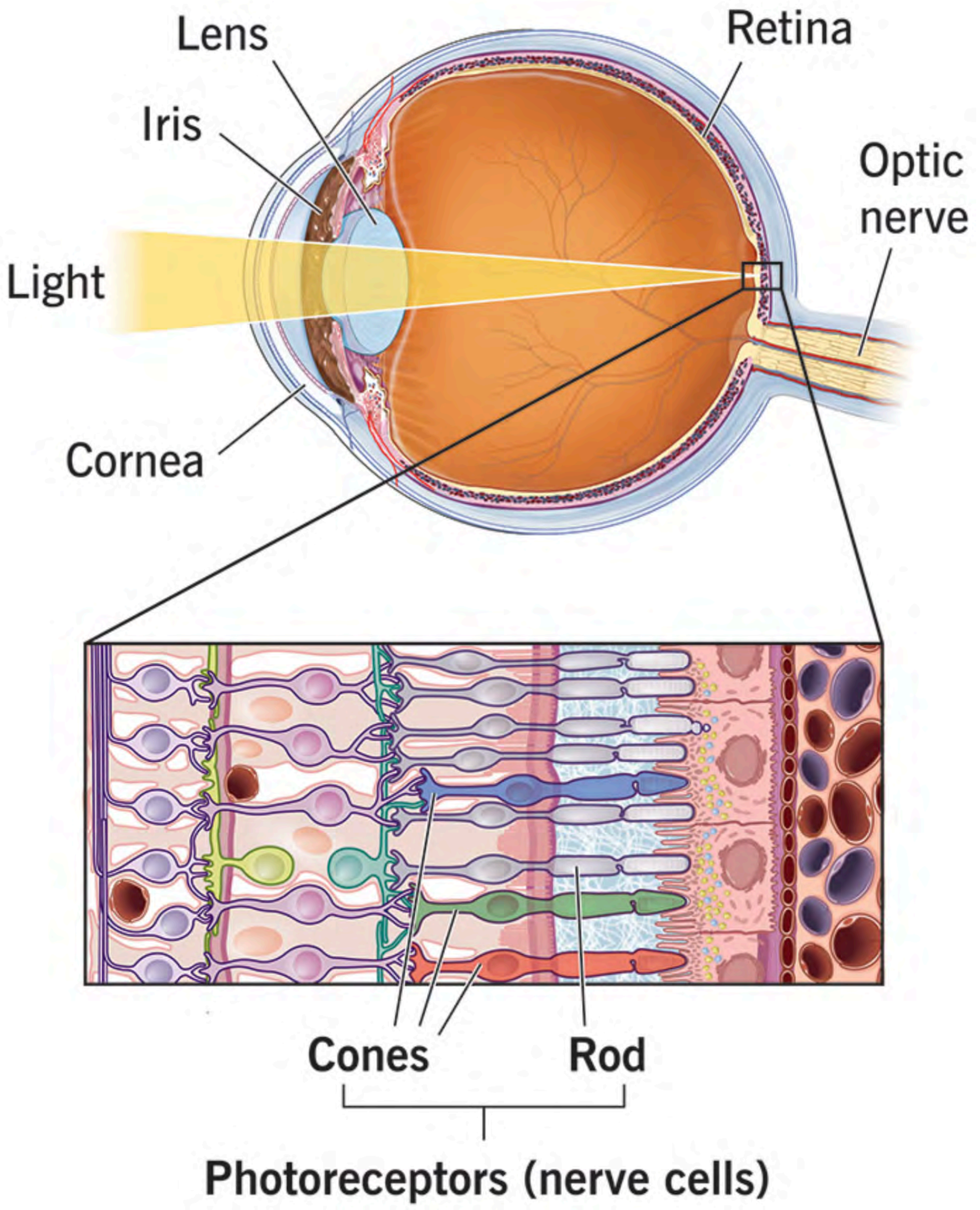
Two main types of sensors:

Rods - No color detection; sensitive to light; night vision

Cones - Three types of cones that are sensitive to red, green, or blue



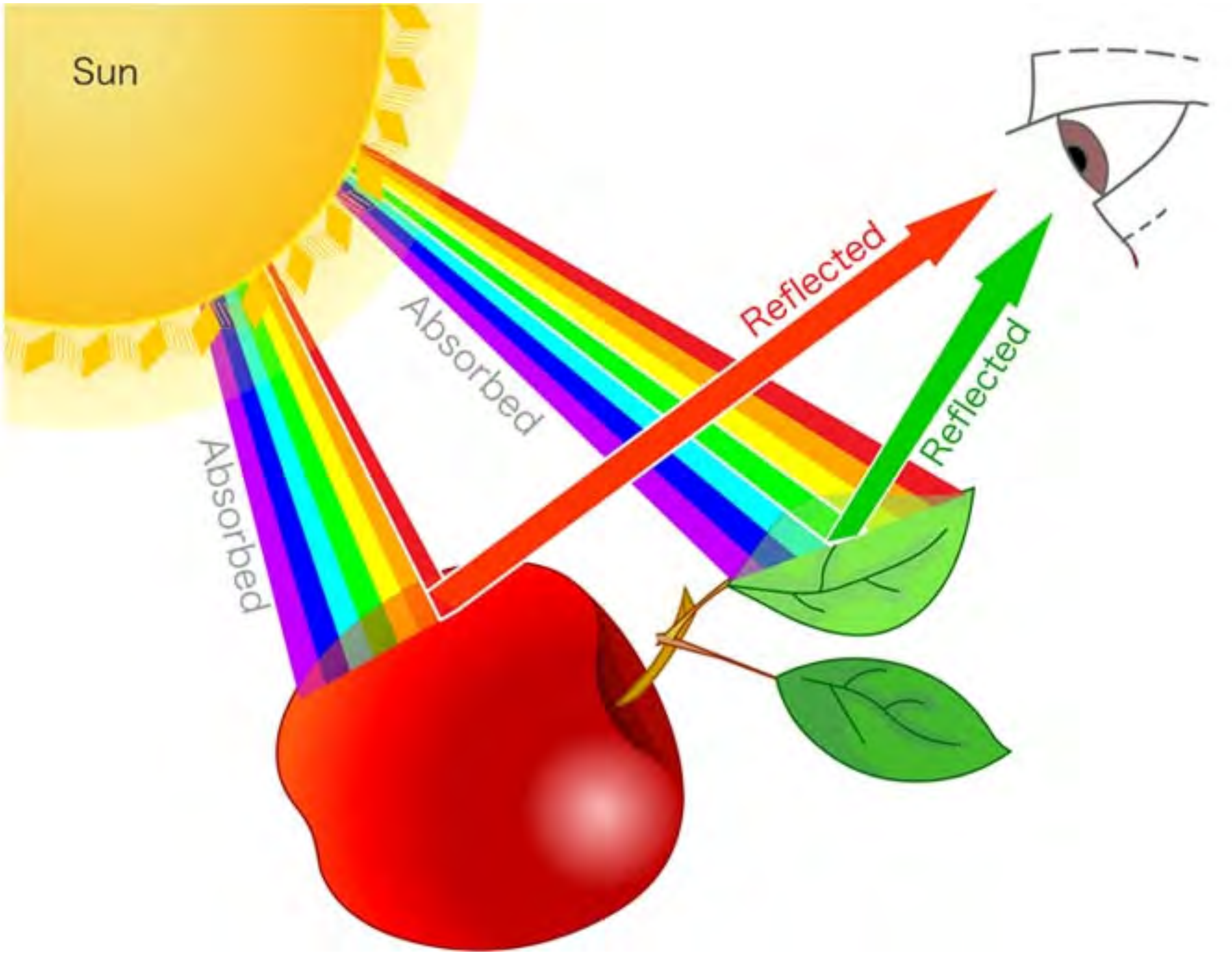
[source: Wikipedia]



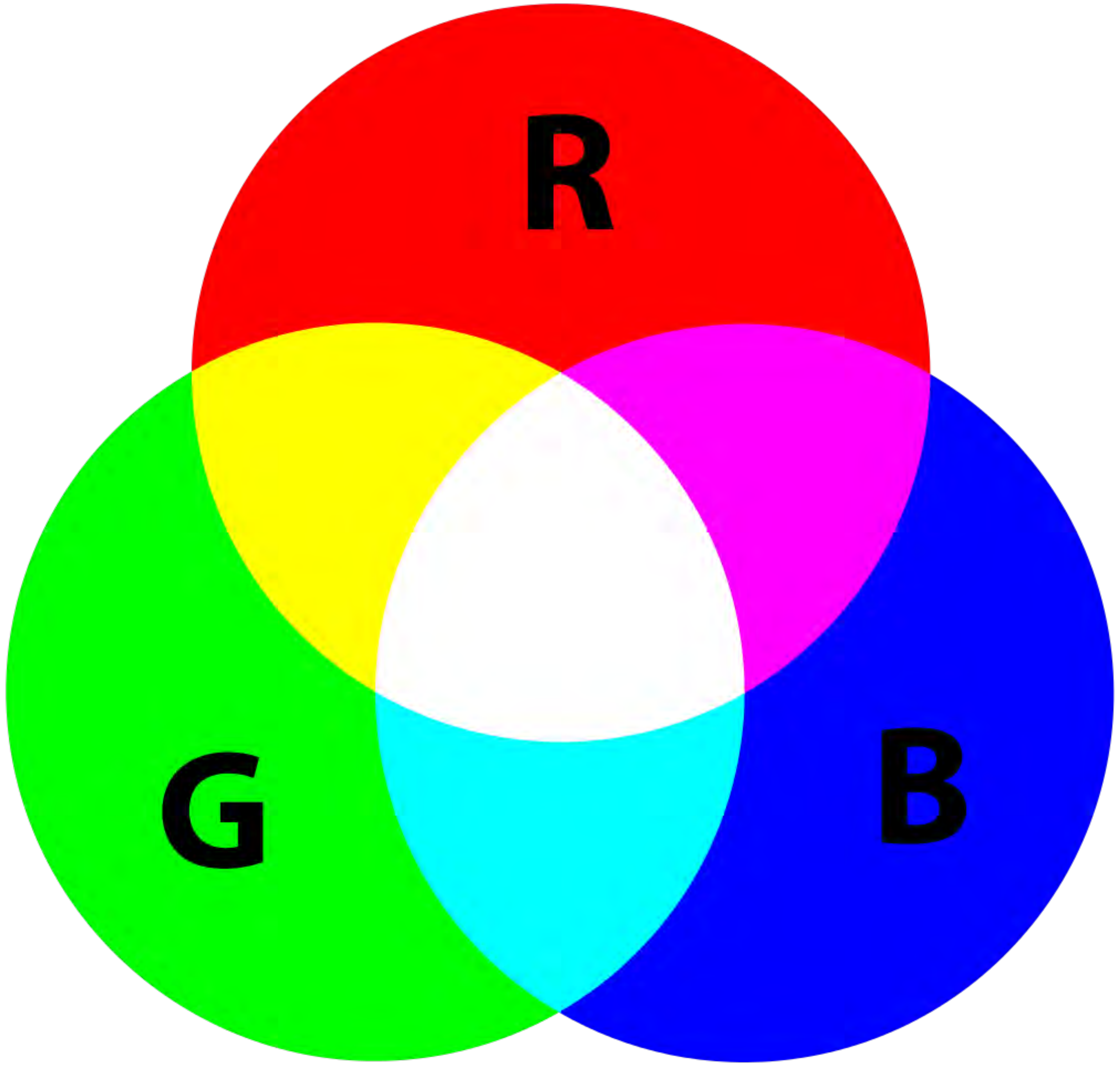
[source: Cleveland Clinic]

RGB Color System

Better for displaying colors on digital screens



[source: Datacolor]



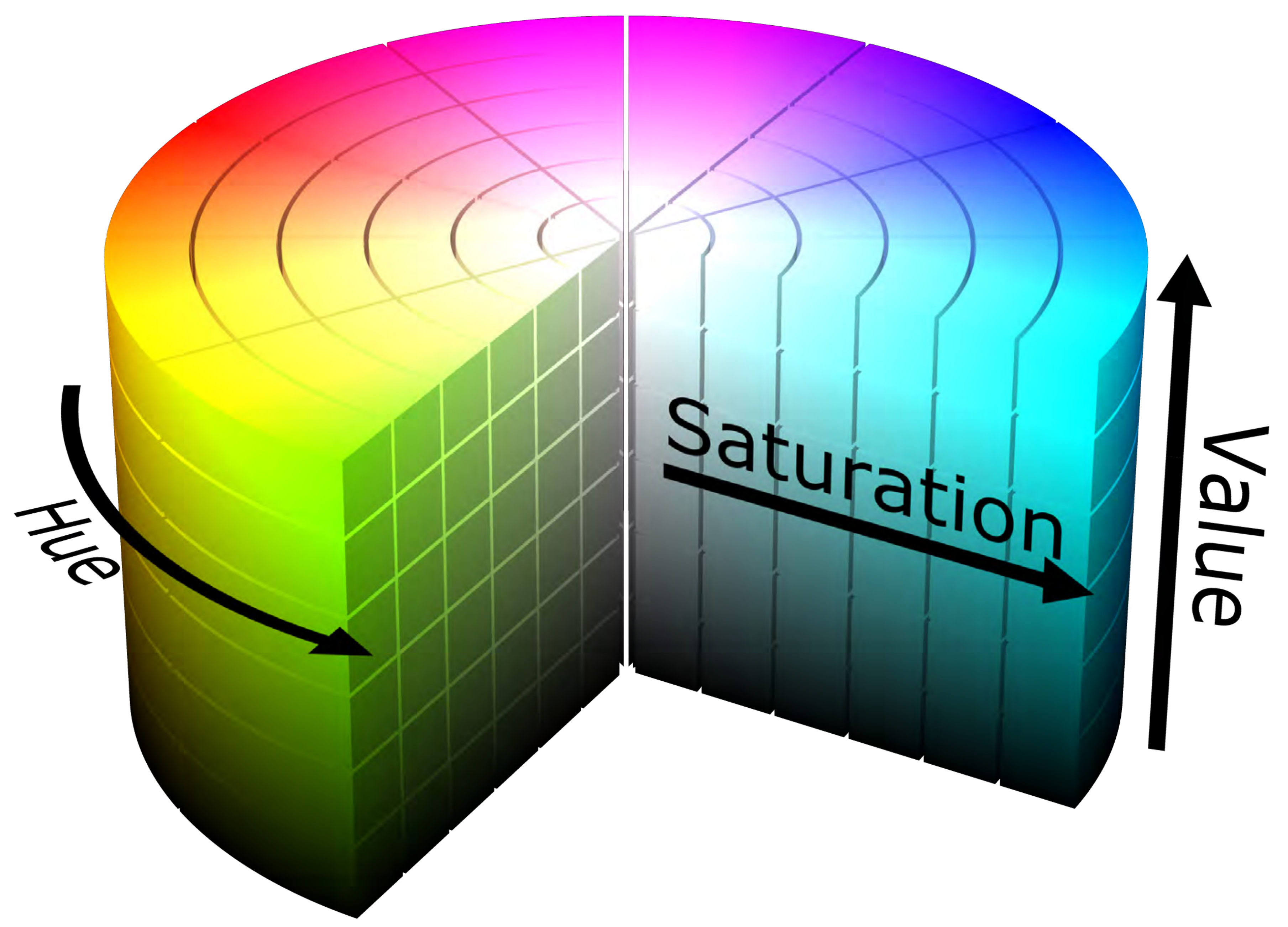
HSV Color System

More human-friendly

Hue - basic color types (red, blue, green, yellow, etc.) , measured as an angle from 0° to 360° on a color wheel.

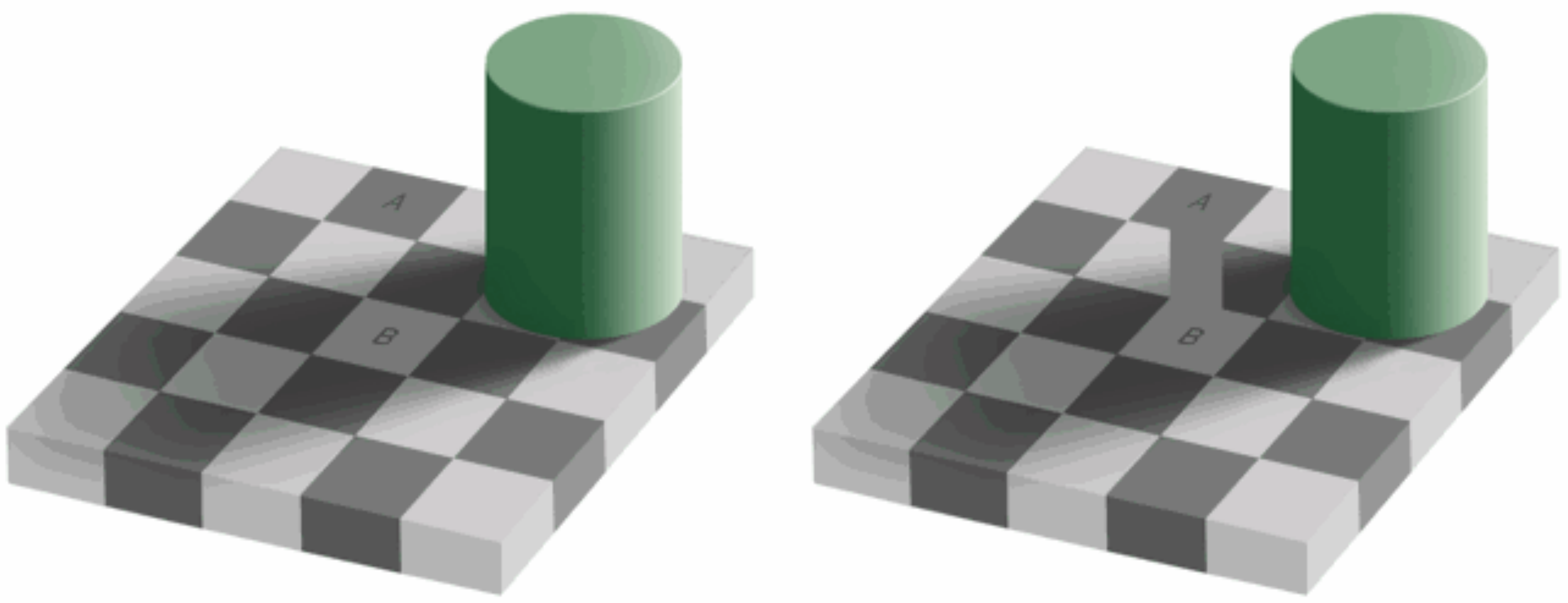
Saturation - The intensity or vibrancy of a color, ranging from 0% (grayscale) to 100% (pure color).

Value - the brightness of the color, ranging from 0% (black at bottom) to 100% (full brightness at top)



Luminance and Color Images

1. Our eyes are more sensitive to brightness change
2. Luminance captures brightness
3. Perception of brightness and color is context-dependent

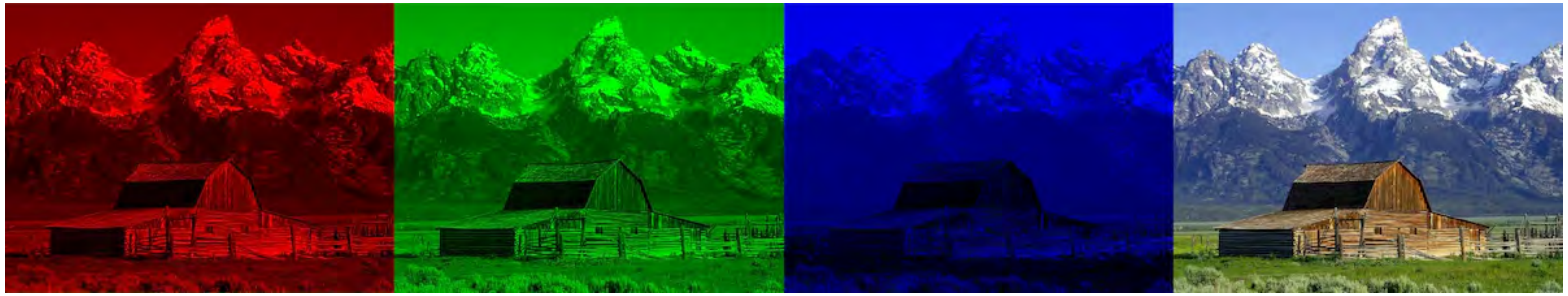


[source: checker shadow illusion by edward adelson]

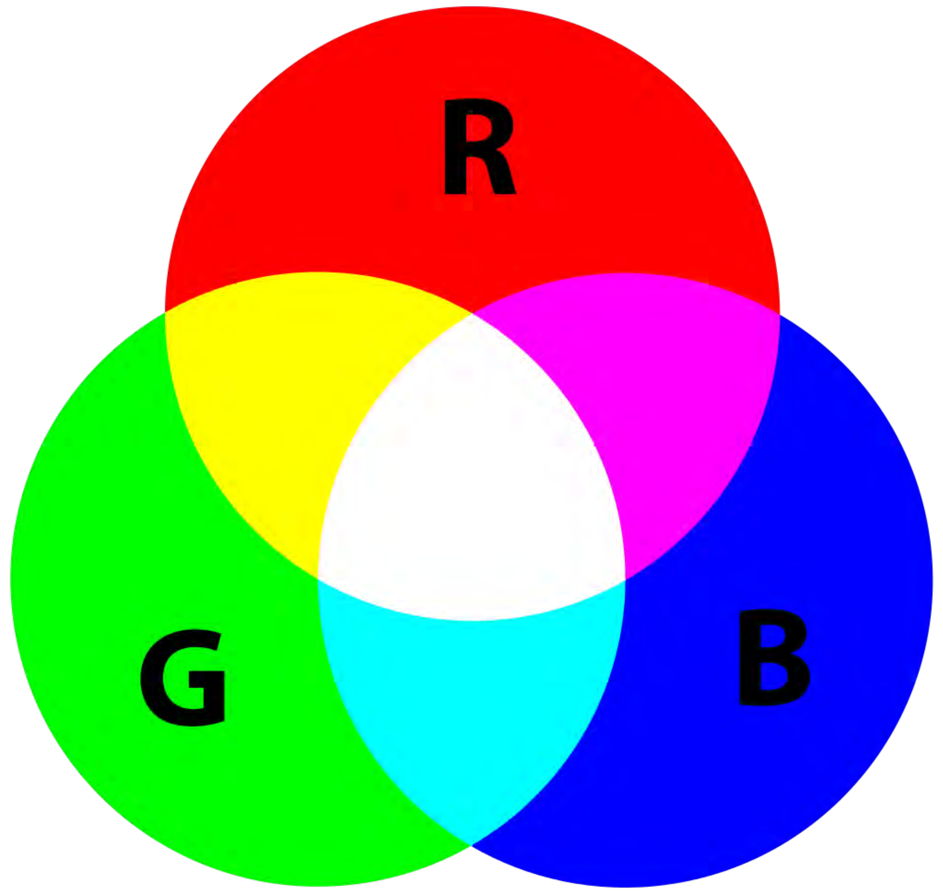
Additive Color

How light produces color

1. Mixing transmitted light
2. Red, Green and Blue light
3. The result is white if we mix all three colors equally
4. For devices that emit light



[source: Dinfos]

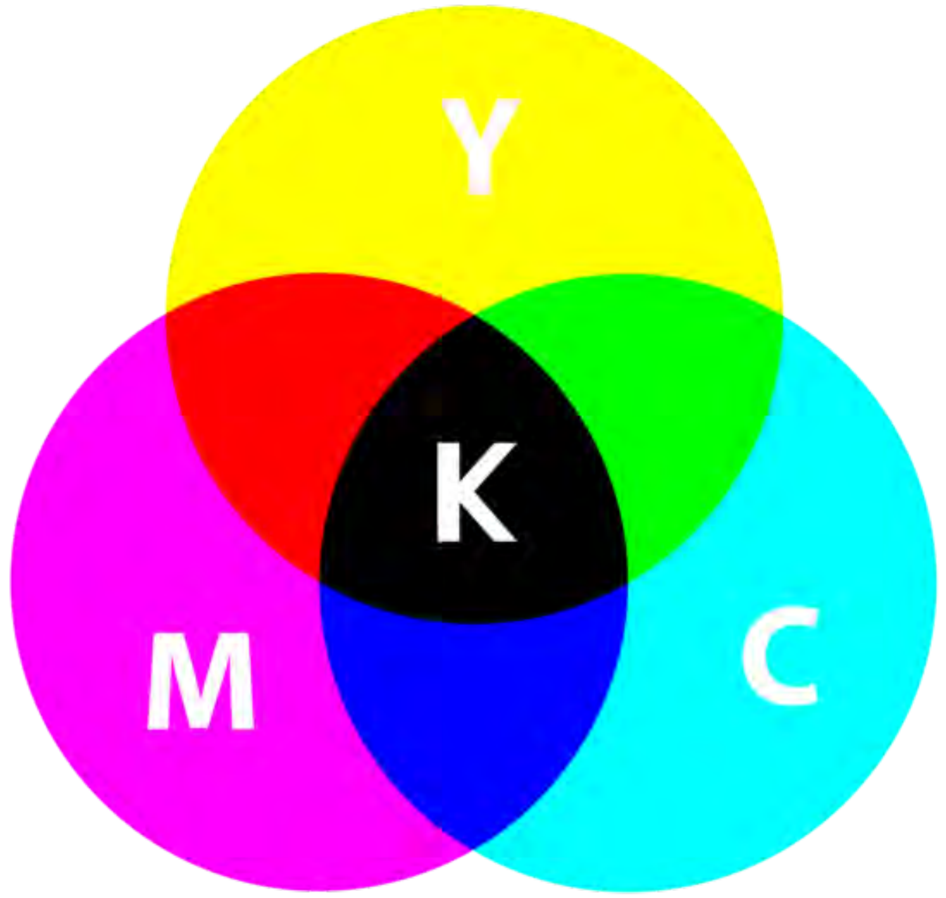


Subtractive Color

Pigments produce color using reflected light

1. Mixing reflected light
2. cyan, yellow, magenta and black pigment
3. The result is black if we mix them together
4. For materials that reflect light

[source: Dinfos]



Gamma Correction

$$Brightness = V^{\gamma}$$

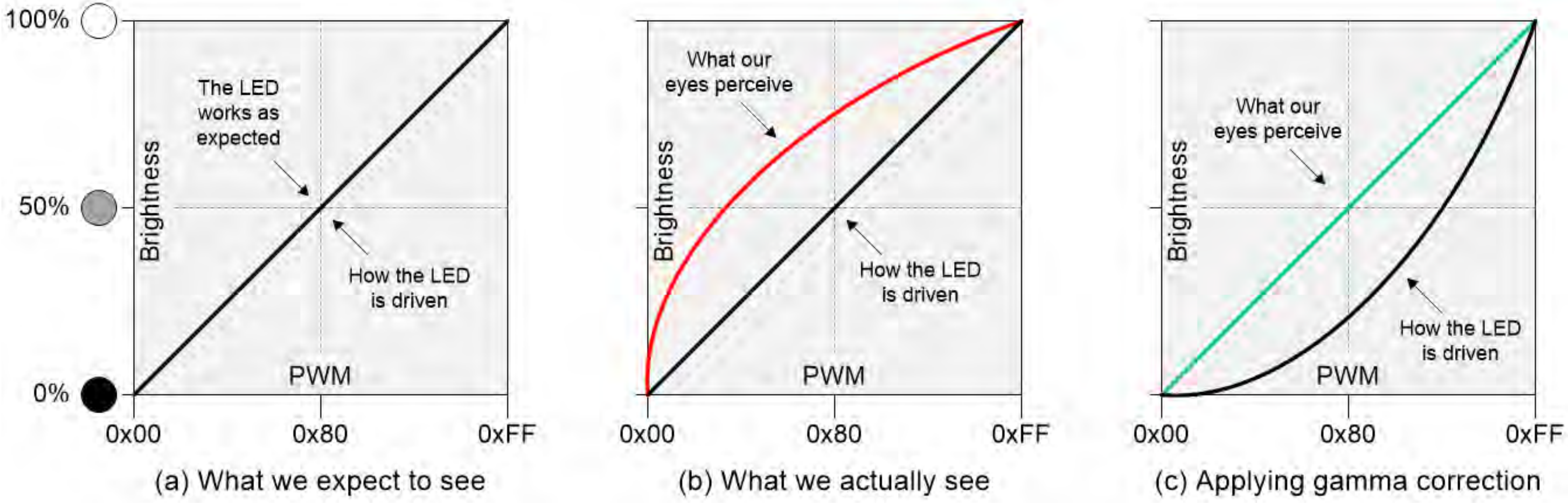
V voltage

Pixel values

$V_{\text{corrected}} = V_{\text{desired}}^{1/\gamma}$ $\quad V_{\text{corrected}}, V_{\text{desired}}$ pixel values from [0, 1]

γ =2.2 for most monitors

Gamma Correction



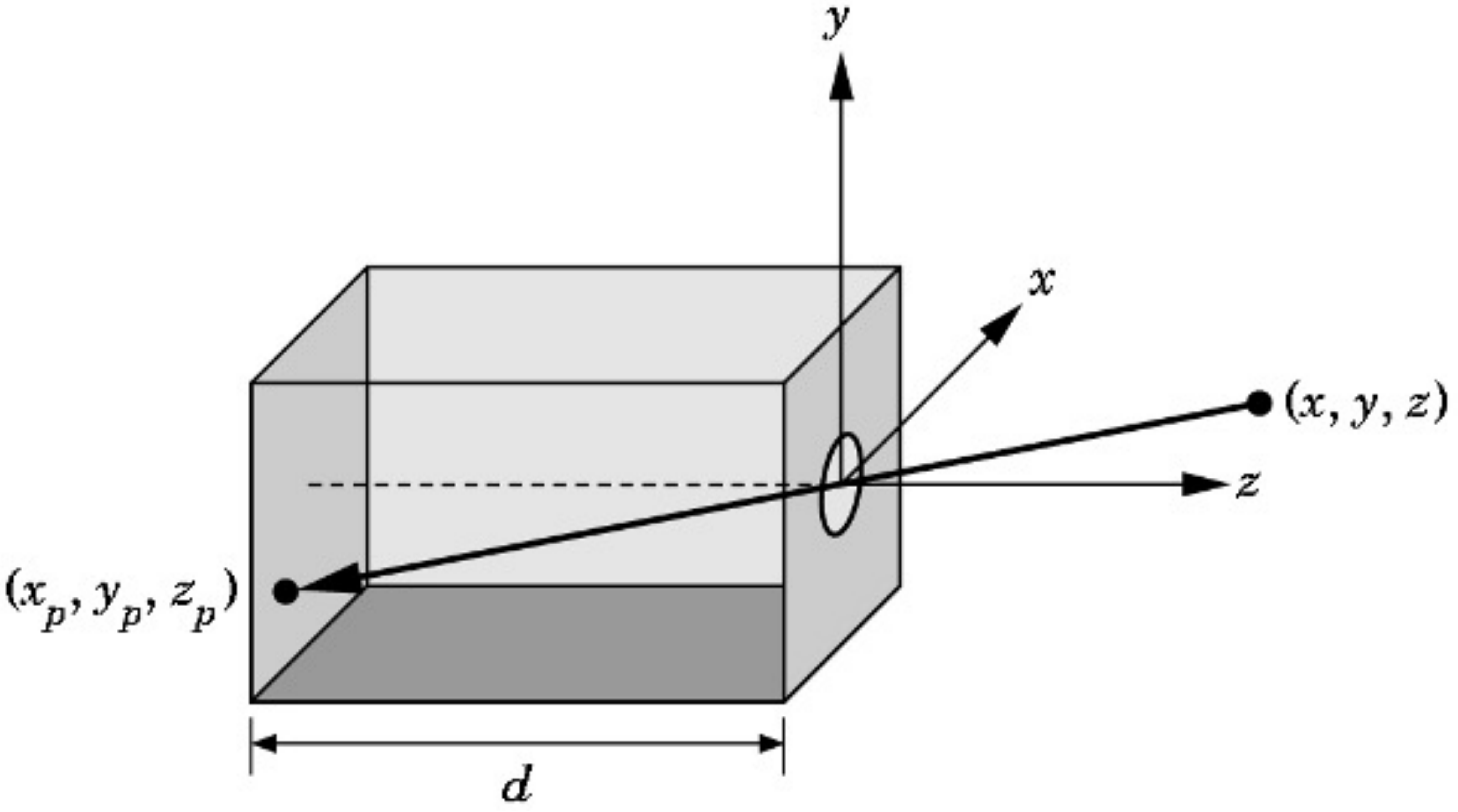
[source: CliveMaxfield]

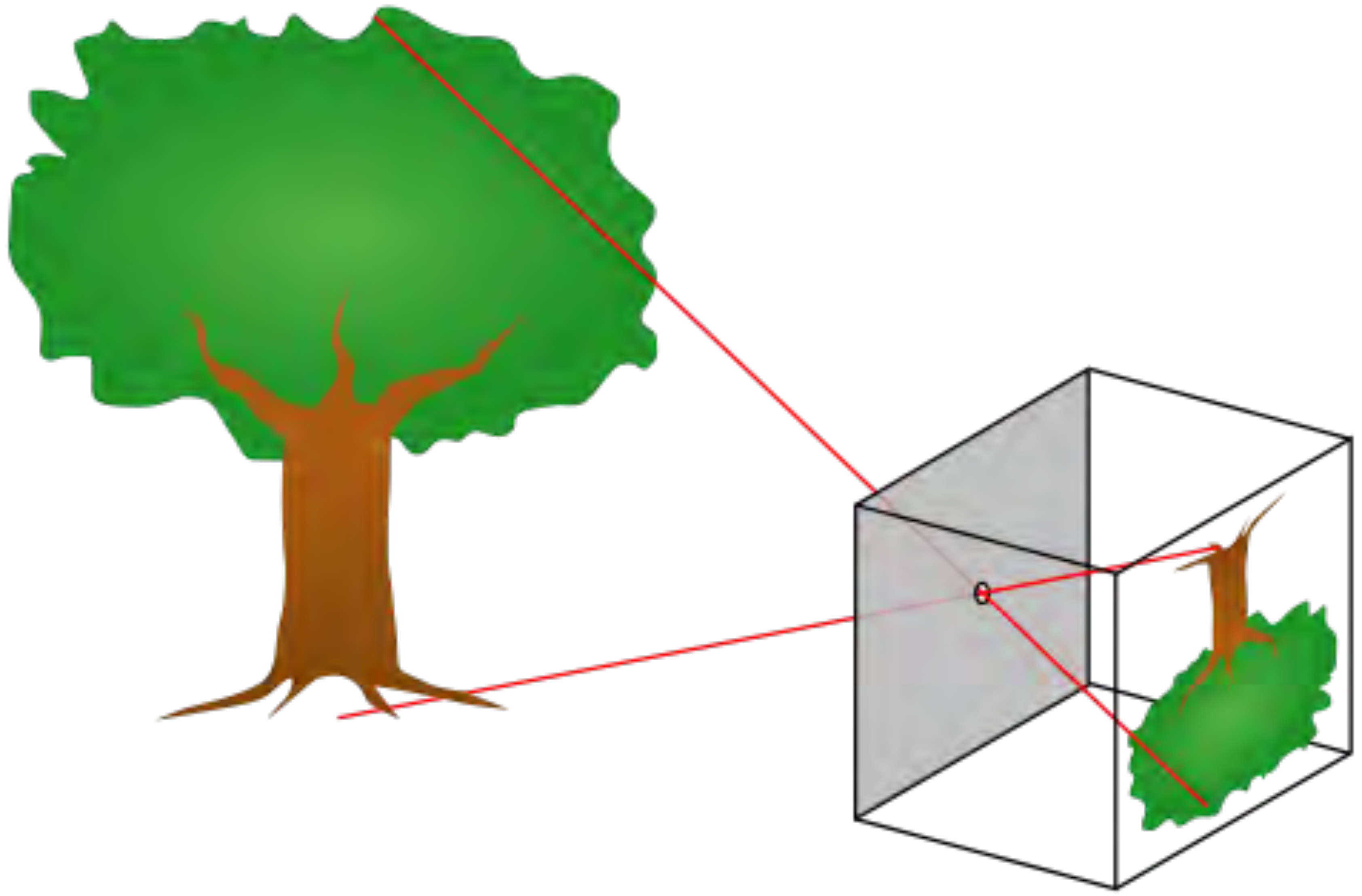
$$V_{\text{out}} = V_{\text{in}}^{1/\gamma}$$

γ =2.2 for most monitors

$$V_{\text{in}} = V_{\text{out}}^{\gamma}$$

Pinhole Camera

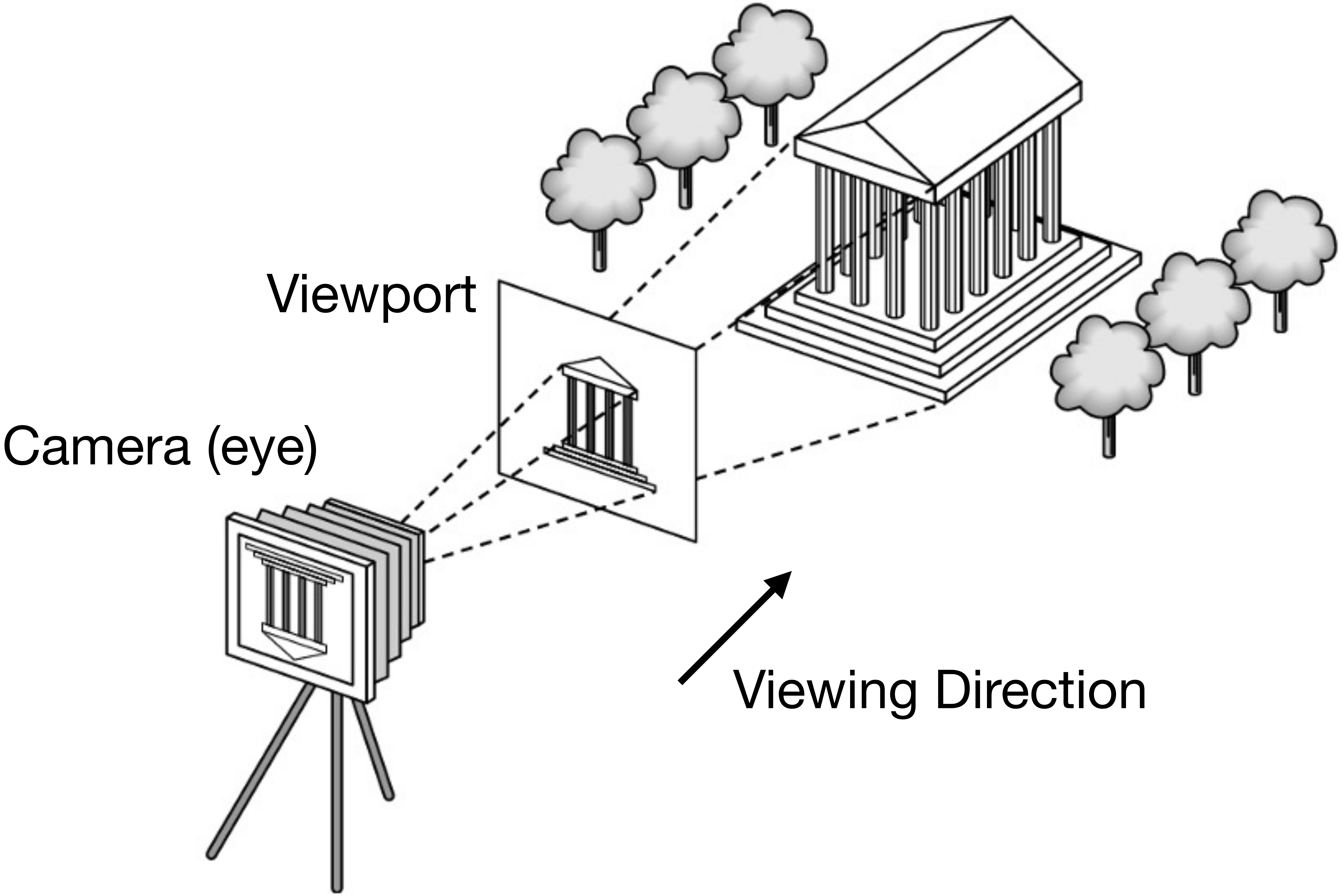




[source: Wikipedia]

$$x_p = -\frac{x}{z} \cdot d \qquad y_p = -\frac{y}{z} \cdot d \qquad z_p = d$$

Synthetic Camera



Model matrix

View matrix

Projection matrix

$$v' = M \cdot V \cdot P \cdot v$$